Paper 5 — REPRESENTATIONS (Quantum): Quantum Structure as Least-Cost Admissible Bookkeeping

Hilbert Space, the Born Rule, and Measurement under Finite Enforceability

E. Brooke

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Abstract

This paper derives the mathematical structure of quantum mechanics — Hilbert space, the Born rule, tensor products, unitarity, and CPTP dynamics — as the realized least-cost admissible bookkeeping for coherent jointly enforceable distinctions under finite capacity. Quantum mechanics is not postulated; it is selected by the admissibility framework of Papers 1–4 in regimes where coherent reversible refinement holds. The derivation proceeds without assuming probability, dynamics, or any quantum postulates. The v2.0-PLEC release reframes the entire derivation around the Principle of Least Enforcement Cost (PLEC): admissibility fixes the licensable class of bookkeeping schemes; PLEC selects the realized minimal scheme; measurement is a change of admissible class rather than a collapse axiom. Imported closure theorems (Solèr, Gleason) are tagged as imports closing an APF-prepared admissible class, not as APF-internal derivations.

Wayfinder

Operational primitives used in this paper.

- **Admissible class $\mathcal{A}_\Gamma$:** the bookkeeping schemes physically licensable in context Γ under A1.
- **Enforcement-cost functional E_Γ:** total capacity burden required to maintain a configuration $X \in \mathcal{A}_\Gamma$.
- **PLEC selection X^*:** realized structure, $X^* \in \arg\min_{X \in \mathcal{A}_\Gamma} E_\Gamma(X)$.
- **Coherent regime:** reversible refinement, recombination, undo all preserve admissibility.
- **Record-locked regime:** capacity irreversibly committed to a logged alternative; algebra extends to $M_{\text{sys}} \otimes Z_R$.

This paper writes the quantum formalism as the realized PLEC selector on the coherent admissible class, and writes measurement as the regime exit from coherent to record-locked admissible bookkeeping.

Shared Canonical Frame (Papers 5–6)

Meaning, Admissibility, Realization, Regime Exit.

1. **Meaning.** A distinction is physically meaningful iff it is robustly enforceable under admissible perturbations.
2. **Admissibility.** A configuration, path, or representation is admissible iff its total enforcement demand does not exceed available capacity.
3. **Realization.** When the admissible class is nonempty and the enforcement-cost functional is well-defined on it, realized physical structure is selected as the least-enforcement-cost member of that admissible class.
4. **Regime exit.** When the conditions required for comparison by smooth finite cost fail, the representation exits its domain, while the deeper admissibility constraints remain in force.

1 Introduction

1.1 The Question

Why does quantum mechanics have the mathematical structure it does? Standard presentations take Hilbert spaces, complex amplitudes, the Born rule, tensor products, and unitary evolution as axioms or postulates — accepted because they work. But the formalism permits vastly more structure than nature realizes. Why the specific subset that is physically realized?

1.2 The Answer

Quantum structure is not arbitrary. It is the least-cost admissible bookkeeping required to track coherent jointly enforceable distinctions under finite enforceability. The admissibility framework of Papers 1–4 determines which bookkeeping structures are physically licensable in coherent regimes; quantum mechanics is the realized minimal structure within that admissible class. In this paper we show that, in bilinear unsaturated regimes, least-cost admissible bookkeeping has the familiar form: linear structure from interference closure, inner products from admissibility-invariant overlap, complex structure from continuous reversible symmetry, tensor products from compositional closure, the Born rule from admissibility-invariant probability assignment, and unitary/CPTP evolution from admissibility-preserving reversible/general transformations. No quantum postulates are assumed. The formalism is selected by admissibility.

1.3 What This Paper Assumes

This paper assumes Papers 1 (SPINE), 2 (STRUCTURE), 3 (LEDGERS), 4 (CONSTRAINTS). It also assumes a bilinear unsaturated regime: the interaction functional I_Γ is approximately bilinear

and no admissibility saturation is encountered. This is a regime condition, not a universal truth.

1.4 What This Paper Does NOT Assume

Hilbert space (derived), complex numbers (derived), Born rule (derived), unitarity (derived), measurement postulates (regime transitions, not postulates). No quantum axioms are assumed.

1.5 Reading Guide

Section 2 introduces the canonical PLEC schema. Sections 3–7 walk through the realized least-cost bookkeeping structure step by step. Section 8 treats measurement as regime exit. Section 9 links the bookkeeping derivation to the matter-content results of Paper 4. Section 11 closes with an explicit epistemic-tag table separating internally derived results [P], imported closures [P]+[I], and structural identifications [P_structural].

2 Canonical PLEC Schema

Definition 2.1 (Admissible bookkeeping class)**.** *Let Γ denote a physical context, interface, or regime. Let $\mathcal{A}_\Gamma$ denote the class of all bookkeeping schemes X that satisfy the APF admissibility conditions in that context.*

Definition 2.2 (Enforcement-cost functional)**.** *Let $E_\Gamma : \mathcal{A}_\Gamma \to [0, \infty)$ assign the total capacity burden required to maintain X in context Γ. The minimal value $\inf_{X \in \mathcal{A}_\Gamma} E_\Gamma(X)$ defines the realized cost in Γ.*

Definition 2.3 (PLEC selection)**.** *A realized bookkeeping structure X^* in context Γ is selected by*

$$X^* \in \operatorname*{arg\,min}_{X \in \mathcal{A}_\Gamma} E_\Gamma(X).$$

When the minimizer is unique up to the relevant equivalence relation $\sim$ (basis, gauge, descriptive coding redundancy), the realized structure is unique up to $\sim$.

Remark 2.4. *The PLEC schema separates what is* physically licensable *from what is* physically realized*. The first is fixed by admissibility; the second is fixed only when the admissible class supports a meaningful least-cost comparison. The remainder of this paper instantiates the schema for coherent unsaturated regimes and shows that the realized bookkeeping is quantum mechanics.*

3 Coherence Defines the Admissible Bookkeeping Class

3.1 Coherent Distinctions

A set of distinctions is *coherent* when it can be reversibly refined, recombined, and undone without violating admissibility. Coherence does more than describe a convenient regime: it defines the

class of bookkeeping structures that are physically licensable in unsaturated reversible settings. A bookkeeping scheme that fails coherence is not merely inconvenient; it does not correctly represent the admissible refinement structure of the underlying distinctions.

3.2 Interference Closure

Interference closure is the central consistency condition on the coherent admissible class. If two refinement paths lead to the same physical situation, bookkeeping must identify them consistently. Coherent bookkeeping is therefore required to close under admissible recombination. Concretely: a distinction d that can be refined two ways — $d \to \{d_1, d_2\}$ and $d \to \{d_1', d_2'\}$ — must yield bookkeeping that is closed under reversible recombination of the two refinements.

3.3 Linearity

Theorem 3.1 (Interference closure forces linear structure)**.** *Within the coherent admissible class, interference closure forces the space of realized bookkeeping states to form a vector space over a division algebra.*

This is the first PLEC selection. Quantum linearity is not postulated in advance; it is the minimal closure property of admissible coherent bookkeeping. Among all licensable closure patterns, only the linear closure satisfies coherent recombination at minimum redundancy.

4 Invariant Overlap and the Selection of Complex Hilbert Structure

4.1 Admissibility-Invariant Overlap

Once coherent bookkeeping is restricted to the admissible class, further structure is fixed by invariance requirements. A similarity measure between coherent realizations must remain invariant under admissible reversible transformations and distribute appropriately over refinements. This forces a sesquilinear form: an inner product. Completing that structure yields a Hilbert space.

4.2 Why Complex Numbers?

Admissibility alone does not license every imaginable Hilbert-like representation. The admissible class is further restricted by continuity of reversible symmetry. By Solèr's theorem, infinite-dimensional orthomodular spaces over a division algebra reduce to $\mathbb{R}, \mathbb{C}, \mathbb{H}$. Physical continuity of one-parameter reversible families in two-dimensional coherent subspaces selects $\mathbb{C}$. The phase $e^{i\theta}$ parameterizes the unique continuous family of admissibility-preserving rotations in a 2D coherent subspace. Solèr is cited as an external import; there is no standalone APF check for it, but the APF-internal preparation of the orthomodular class is the ST-chain in `check_L_ST_algebra` (`core.py`) and its successors (`check_L_ST_Hilbert`, `check_L_ST_Dirac`).

Bridge

The role of Solèr here is precisely a closure theorem on an APF-prepared admissible class. APF supplies the orthomodular structure, the division-algebra structure, and the continuity requirement on one-parameter reversible families; Solèr then closes the classification to $\mathbb{R}, \mathbb{C}, \mathbb{H}$; continuous symmetry selects $\mathbb{C}$. Tag: [P] + [I: Solèr].

4.3 Hilbert Space

Completing the inner-product space yields a Hilbert space $\mathcal{H}$. This is the realized least-cost admissible representation of coherent distinction structure in regimes where reversible refinement and continuous symmetry both hold. The complex Hilbert space is therefore not an arbitrary mathematical preference; it is the minimum-redundancy bookkeeping consistent with admissible coherent refinement.

5 Composition and the Realized Composite Structure

5.1 Joint Enforceability is Conditional

Papers 1–2 establish that composition can fail: not every pair of systems is jointly enforceable. This means the admissible class of composite bookkeeping structures is already conditional. When two systems are jointly enforceable, bookkeeping must represent that success of composition without violating refinement consistency.

5.2 Bilinear Composition Forces the Tensor Product

Let A and B be coherent systems with Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$. If they are jointly enforceable, refinement composability requires that refining A then composing with B gives the same result as composing then refining. Bilinearity follows. The minimal admissible composite bookkeeping satisfying bilinear composition and closure under admissible recombination is the tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$.

5.3 Tensor Products are Realized, Not Universal

Tensor products are not universally primitive. They are the realized least-cost bookkeeping for composition *within the admissible region*. Outside that region — at saturation, measurement, or horizons — the tensor-product representation itself ceases to apply. Quantum composition is therefore conditional at the same level as physical joint enforceability.

5.4 Entanglement

Most states in $\mathcal{H}_A \otimes \mathcal{H}_B$ are not product states. These are entangled states: joint realizations that cannot be factored into independent realizations. Entanglement is not a mystery in this framework. It is the generic representation of joint enforceability when the interaction functional is nonzero; entangled states encode correlations that require coordinated bookkeeping across interfaces.

5.5 Conditional Composition (Paper 2 Link)

The tensor product exists *if and only if* the composite is admissible: $A \otimes B$ exists $\iff E_\Gamma(A \cup B) \leq C_\Gamma$. This is non-closure at the level of quantum representations: the familiar assumption that composition always works is valid only within the admissible region. At saturation boundaries, the tensor-product description itself breaks down and the regime exits.

6 The Born Rule as the Unique Admissible Probability Assignment

6.1 Refinement Invariance

Probability assignments must be refinement-invariant: coarse-graining and fine-graining must agree. If outcome E can be refined into $E_1 \cup E_2$ with $E_1 \cap E_2 = \emptyset$, then $p(E) = p(E_1) + p(E_2)$. This is additivity over disjoint outcomes.

6.2 Admissibility Invariance

Probability assignments must be admissibility-invariant: they must not depend on arbitrary bookkeeping choices. For a state ρ and effect E, $p(E \mid \rho)$ must be invariant under basis changes, equivalent representations, and admissible reversible transformations.

6.3 Gleason's Theorem

Theorem 6.1 (Gleason 1957, as a closure on the admissible class)**.** *In a complex Hilbert space of dimension ≥ 3, the unique probability measure satisfying refinement additivity and admissibility-invariance is*

$$p(E \mid \rho) = \mathrm{Tr}(\rho E).$$

For pure states $\rho = |\psi\rangle\langle\psi|$ and projections $E = |\phi\rangle\langle\phi|$, this gives $p = |\langle\psi \mid \phi\rangle|^2$.

APF-internal verification of the finite-dimensional case is in `check_L_Gleason_finite` (`supplements.py`); the full Gleason statement remains an external import.

Bridge

The important point is not merely that Gleason's theorem applies, but *why* it applies here. APF does not conjure probability from nowhere; it identifies the admissible class of bookkeeping structures and the invariance conditions any physically meaningful assignment must satisfy. The Born rule is then the unique probability assignment compatible with that realized bookkeeping. Tag: [P] + [I: Gleason 1957].

6.4 The Born Rule Is Selected, Not Postulated

The Born rule is not a postulate. It is the unique admissibility-invariant probability assignment in the coherent admissible class once Hilbert structure is selected. The "mystery" of why probability is $|\psi|^2$ dissolves: it is the only assignment compatible with the bookkeeping forced by admissibility.

7 Dynamics as Admissibility-Preserving Evolution

7.1 Reversible Evolution: Unitarity

Theorem 7.1 (Admissibility-preserving reversible evolution is unitary)**.** *Within the realized coherent bookkeeping structure, reversible evolution must preserve admissibility. Preservation of the inner product forces unitary or antiunitary transformations; physical continuity excludes antiunitary maps except for discrete cases such as time reversal. Continuous reversible evolution is therefore generated by self-adjoint operators: $U(t) = e^{-iHt}$. This is the Schrödinger structure.*

7.2 General Evolution: CPTP Maps

Not all evolution is reversible. Record formation, environmental interaction, and capacity commitment produce irreversible changes. Admissibility-preserving transformations that remain admissibility-preserving under extension by ancillas are completely positive. With normalization (trace preservation), admissibility-preserving evolution is described by CPTP maps.

7.3 Dynamics is Not a Separate Postulate

The familiar dynamical structure of quantum mechanics is not a separate postulate layered on top of Hilbert space. It is the admissibility-preserving evolution law of the realized coherent bookkeeping selected in the preceding sections.

8 Measurement as Exit from the Coherent Bookkeeping Regime

8.1 The Measurement Problem Re-Read

Measurement is the central conceptual difficulty of standard quantum mechanics because the formalism appears to require an extra collapse postulate that interrupts otherwise unitary evolution. In the admissibility framework, the issue is reformulated more cleanly. The question is not how a unitary system suddenly violates its own law, but *when the coherent bookkeeping regime ceases to be the correct admissible representation.* Measurement is therefore a regime-transition problem, not an axiom problem.

8.2 Before and After Measurement as Class Change

Before measurement, multiple outcomes are jointly admissible and the relevant bookkeeping class is the coherent class: reversible refinement, recombination, and admissibility-invariant overlap are all meaningful. After measurement, capacity has been irreversibly committed to a recorded alternative. The relevant bookkeeping class is no longer the coherent one; it is the record-locked class in which mutually incompatible alternatives are excluded by the structure of the irreversible log.

The key point is that this is a *change of admissible class.* Quantum superposition does not "mysteriously disappear." Rather, the system exits the coherent least-cost bookkeeping regime and enters a record-locked regime whose admissible structure is different.

8.3 Record-Locking Channels as Explicit Algebraic Class Change (T9 / L3-μ)

The record-locking channel formalization makes this change explicit. The interface algebra decomposes as

$$M = M_{\text{sys}} \otimes Z_R,$$

where M_{sys} is the coherent system algebra and Z_R is a classical log algebra with projections $\{p_s\}$ indexed by log strings. Each enforcement operation $\mathcal{E}_i$ acts both on the system and through an irreversible append map $\alpha_i : p_s \mapsto p_{(s,i)}$ on the log:

$$\mathcal{E}_i = (\mathcal{E}_i^{\text{sys}} \otimes \text{id}) \circ (\text{id} \otimes \alpha_i).$$

Theorem 8.1 (T9 / L3-μ, $k!$ inequivalent histories)**.** *For k enforcement operations applied in all $k!$ orderings, the resulting CP maps are pairwise distinct: for $\pi \neq \sigma$, $\mathcal{E}_\pi \neq \mathcal{E}_\sigma$. The log projections p_π and p_σ are orthogonal. Therefore $|H_k/\equiv| = k!$ inequivalent histories.*

Formal verification: `check_T9` (`gauge.py`).

This construction does more than reinterpret measurement. It exhibits the new admissible class algebraically. What is traditionally called collapse is the point at which the physically relevant admissible description is no longer supported on M_{sys} alone, but on $M_{\text{sys}} \otimes Z_R$ with irreversible sector separation. The log is irreversible because record-lock capacity constraints prevent undoing the append (T0.4′ certificate, Paper 2).

8.4 Decoherence as Progressive Class Transfer

Decoherence is the progressive transfer from coherent admissible bookkeeping to record-locked admissible bookkeeping. As environmental records accumulate, the log algebra Z_R grows and the coherent sector becomes a less accurate effective compression of the admissible structure. When enough capacity has been committed to records, the quantum description ceases to apply — not because it is false, but because the system is no longer in the regime whose admissible class quantum mechanics represents.

Bridge

Measurement is to Paper 5 what saturation, branching, and horizons are to Paper 6: not a violation of deeper law, but a boundary of representational validity. The unifying project-level statement is: record locking, saturation, horizons, and topology change are all regime exits of representation, not failures of admissibility itself.

9 Connection to Matter Content

9.1 Quantum Structure Constrains Particle Content

The quantum representation derived in this paper does not float freely above the constraint structure. It is jointly constrained with the gauge and matter content derived in Papers 1 and 4 by the same capacity architecture:

Quantum structure	Capacity constraint	Consequence	Coderef
$\mathcal{H}_A \otimes \mathcal{H}_B$	Conditional on admissibility	Composition fails at horizons	—
Unitary symmetry group	Gauge selection	$SU(3)\times SU(2)\times U(1)$	`check_T_gauge` (`gauge`
Matter representations	Anomaly cancellation	Chiral fermion content forced	`check_T5` (`gauge.py`)
Generation structure	Capacity saturation	$N_{\text{gen}} = 3$	`check_T4F` (`gauge.py`)
Born-rule probabilities	Gleason + admissibility	$p = \lvert\langle\psi \mid \phi\rangle\rvert^2$	`check_L_Gleason_fin`

The quantum representation is not free to have arbitrary matter content. The same capacity architecture that forces Hilbert space also forces three generations of chiral fermions in the Standard Model gauge group. This is qualitatively new: prior derivations of quantum structure (Hardy, Chiribella et al.) do not constrain particle content.

9.2 Particles Require Quantum Structure

The enforcement potential $V(\Phi)$ derived in Paper 3 (Section 5.3) shows that particle-like excitations require the quantum structure derived here. The binding well at $\Phi/C \approx 0.81$ has mass gap $d^2V/d\Phi^2 > 0$, but no classical soliton localizes: particles are quantum modes of the enforcement

field, not classical cost minima. This closes the conceptual gap between the constraint architecture and observable particle physics.

9.3 Bridge Tightening (Open)

The bridge from bookkeeping structure to gauge and matter content is presently a structural coexistence claim under common capacity architecture, not a direct single-theorem derivation. Tightening the bridge from compatibility to realized selection is one of the open theorem projects identified in the v2.0-PLEC revision program.

10 Epistemic Tagging Discipline (Paper 5)

Project-wide categories.

- **[P] Internally derived.** Result follows from APF axioms plus explicitly stated regime assumptions, without external classification.
- **[P]+[I] Derived with import.** APF fixes the admissible class; an external mathematical theorem closes classification, uniqueness, or representation.
- **[P_structural] Structural identification.** The framework identifies an already-derived structure with a familiar physical role; not a stronger derivation than the identification supports.
- **[Pred] Phenomenological proposal/prediction.** Empirical identification or scaling whose status is downstream and must be evaluated against data.

Item	Tag	Coderef
Linear structure	[P]	—
Inner product	[P]	—
Complex field selection	[P]+[I]	`check_L_ST_Hilbert` (`core.py`)
Hilbert completion	[P]	`check_L_ST_Hilbert` (`core.py`)
Tensor product	[P]	—
Born rule	[P]+[I]	`check_L_Gleason_finite` (`supplements.py`)
Unitarity	[P]	—
CPTP maps	[P]	—
Record-locking, $k!$ histories	[P]	`check_T9` (`gauge.py`)
Gauge coupling	[P]	`check_T_gauge` (`gauge.py`)
Generation count	[P]	`check_T4F` (`gauge.py`)
Anomaly cancellation	[P]	`check_T5` (`gauge.py`)
Measurement as class change	[P_structural]	`check_Regime_exit_Type_III` (`plec.py`), `check_T9` (`gauge.py`)
Matter-content bridge	mixed	multiple

This discipline makes the paper easier to defend: imported steps remain visible, while internally derived structure becomes sharper rather than weaker.

11 Summary

11.1 What Is Derived

The realized least-cost admissible bookkeeping for coherent jointly enforceable distinctions in bilinear unsaturated regimes:

- Linear vector-space structure from interference closure.
- Inner-product structure from admissibility-invariant overlap.
- Complex Hilbert space via Solèr + continuous symmetry selection.
- Tensor-product composition (conditional on joint admissibility).
- The Born rule via Gleason on the prepared admissible class.
- Unitary and CPTP evolution as admissibility preservation.
- Record-locking and the $k!$ history sectors of T9 / L3-μ.

11.2 What Is NOT Derived

Specific Hamiltonians (depend on physical content). Coupling constants beyond the structural ratios (regime parameters). Particle masses beyond the structural hierarchy (Yukawa ratios are regime parameters). Particle identification of dark matter (Paper 6 fences this).

11.3 The Status of Quantum Mechanics

Quantum mechanics is the realized least-cost bookkeeping for coherent jointly enforceable distinctions in bilinear unsaturated regimes. It is exact within that regime, conditional on its regime assumptions, and non-fundamental in the sense that it is a representation rather than the ontological base layer. When the conditions of coherent reversible bookkeeping fail — at saturation, record locking, measurement, or horizons — the quantum representation exits its domain, while the deeper admissibility constraints remain in force.

Appendix A: Regime Assumptions and Where They Fail

Assumption	Content	Where it fails
Interference closure	Refine / recombine / undo preserve admissibility	Record-locked hi
Overlap invariance	Similarity invariant under reversible transforms	Near saturation l
Phase admissibility	Continuous one-parameter reversible families exist in 2D subspaces	Discrete / quanti
Compositional closure	Refinement and composition commute for jointly enforceable systems	Saturation; horiz
Bilinearity	I_Γ approximately bilinear	Strong-coupling;

Where these regime assumptions fail, quantum mechanics ceases to apply — but the underlying admissibility constraints (Papers 1–4) continue to hold. The constraint structure is more fundamental than the quantum representation.

Appendix B: Record-Locking Channel Details

The T9 / L3-μ proof constructs the record-locking model explicitly:

1. **Interface algebra.** $M = M_{\text{sys}} \otimes Z_{\leq k}$, where $Z_{\leq k}$ is the classical log algebra with projections $\{p_s\}$ for $s \in \Sigma^m$ ($m \leq k$), $\Sigma = \{1, \ldots, k\}$.
2. **Append maps.** $\alpha_i(p_s) = p_{(s,i)}$ for $|s| < k$; $\alpha_i(p_s) = p_s$ for $|s| = k$ (saturation). Key property: $\alpha_i(p_s) \cdot \alpha_j(p_s) = 0$ for $i \neq j$ (ranges disjoint).
3. **Enforcement channels.** $\mathcal{E}_i = (\mathcal{E}_i^{\text{sys}} \otimes \text{id}) \circ (\text{id} \otimes \alpha_i)$. Applying $\mathcal{E}_i$ refines the system and appends "i" to the irreversible log.
4. **Permutations.** For permutation π, $\mathcal{E}_{\pi(1)} \circ \cdots \circ \mathcal{E}_{\pi(k)}$ maps $1_{\text{sys}} \otimes p_\emptyset$ to $1_{\text{sys}} \otimes p_{(\pi(1),\ldots,\pi(k))} = 1_{\text{sys}} \otimes p_\pi$.
5. **Distinctness.** For $\pi \neq \sigma$, $p_\pi \neq p_\sigma$ and $p_\pi \cdot p_\sigma = 0$. Therefore $|H_k/\equiv| = k!$.

Corollary. $\dim(Z_R) \geq k!$ and $\Delta S_{\text{records}} \geq \log(k!) \sim k \log k$. The entropy of measurement is structural, not thermodynamic.

Appendix C: Changelog

- **v2.0-PLEC (April 2026).** Reframed entire derivation around PLEC schema: admissibility fixes the licensable bookkeeping class, PLEC selects the realized minimum, measurement is regime exit from coherent to record-locked admissible class. Added shared canonical box, wayfinder box, canonical PLEC schema (definitions), epistemic tagging discipline section. Reformatted bridge boxes around Solèr and Gleason as imports closing an APF-prepared admissible class. Cleaned formatting throughout (v1.0 had literal-escape artifacts in math mode and broken section spacing).
- **v1.0 (February 2026).** Record-locking channels (T9 / L3-μ). Decoherence as progressive capacity commitment. Matter-content connection (quantum structure jointly constrains particle content). Regime-assumptions appendix. Forward references to Paper 7 (ACTION) and Paper 4 §7 ($\sin^2\theta_W$).